

FPGA Design Using VHDL

Module 7E - State Encoding

State Encoding I

```
type statetype is (idle, init, check, slowchg, fastchg,
donechg);
signal curstate,next_state : statetype;
```

- **Sequential** - Encodes the states as a binary number
  - 000,001,010,011,100,101
- **One-Hot** - One flip-flop is active, or hot, at any one time.
  - Good alternative when trying to optimize speed or reduce power dissipation
  - 100000, 010000, 001000, 000100, 000010, 000001
- **Auto** – Allows Vivado to determine the best encoding method for a given FSM
  - This is the default option

State Encoding II

- **Gray** - Guarantee only one state variable switches between two consecutive states.
  - Appropriate for controllers exhibiting long paths without branching
  - Minimizes hazards and glitches.
  - 000,001,011,010,110,111
- Many others….
  - For a complete list, see the section on Finite State Machines in the Vivado User Guide
- How do you select FSM encoding?
  - Select Project Manager
  - Click "Settings"
  - Select Synthesis
  - Under settings, -fsm_extraction defines the encoding type

